

Assignment 14 – Robot Framework Variables and Data-Driven Automation

1. Module Name

Robot Framework

2. Experiment Title

Assignment 2: Variables and Data-Driven Automation

3. Problem Statement

Implement a Robot Framework test case that uses variables for reusable test information and reads multiple username and password combinations from an external data file to execute the same automation workflow.

4. Objective

Demonstrate Robot Framework variables, external CSV-based test data, DataDriver integration and reusable test templates for executing multiple positive and negative login scenarios.

5. Tools, Software and Concepts Used

Python; Robot Framework; SeleniumLibrary; Selenium WebDriver; DataDriver library; Google Chrome; Robot Framework Variables; Test Template; CSV test data; IF/ELSE conditional validation.

6. Source Code / Implementation Steps

Defined application URL and browser as Robot Framework variables. Added DataDriver to read login_data.csv. Created a Test Template so the same keyword workflow is executed for every CSV row. The reusable keyword opens SauceDemo, enters the supplied username and password, clicks Login, and validates either the Products page for a successful login or the error element for a failed login.

7. Output / Labelled Screenshots

Screenshot 1 – Robot Data-Driven Execution: terminal output showing all generated test cases completed successfully. The screenshot is stored in Source_Code/Screenshots.

8. Result

The Robot Framework data-driven automation was implemented successfully. Four login data sets are executed using the same reusable test workflow, with positive and negative outcomes validated.

9. Observation

Variables make commonly used values easier to maintain, while external CSV data separates test data from automation logic. DataDriver and Test Template reduce duplication and support scalable test execution.

10. Conclusion

The experiment successfully demonstrated Robot Framework variables and external-file-based data-driven automation using SeleniumLibrary and DataDriver.